

Sensitivity of Series System Reliability Estimation to the Coarsening Conditions on Masked Failure Data

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Abstract

Maximum likelihood estimation for series systems with masked failure data rests on three coarsening conditions: (C1) the failed component is in the candidate set; (C2) the candidate set is independent of the true cause given the failure time; and (C3) the masking mechanism does not depend on the component reliability parameters. None of the three is verifiable from observed data, and any of them can fail in practice. We develop a unified sensitivity-analysis framework for the standard C1-C2-C3 MLE under calibrated violations of each condition. Closed-form pseudo-true expansions yield local sensitivity indices that specialize the Index-of-Local-Sensitivity-to-Nonignorability of Troxel et al. (2004) to set-valued candidate-set coarsening, and an exact total-hazard preservation result via a profile-likelihood factorization sharpens the exponential case from first-order to all-orders. A systematic sweep across all three violations on a 5-component Weibull system validates the resulting robustness hierarchy: system-level estimands (total hazard, mean time to failure, system reliability at a fixed time) are robust under any of the three violations, with system-hazard relative bias bounded under 4 percent throughout, while component-level estimands degrade with violation severity. Scale parameters of low-rate components exhibit the most catastrophic degradation (relative bias exceeding 100 percent under C3, and confidence-interval coverage collapsing to zero for some parameters), so component-level claims should be reported with the first-order robustness intervals introduced here. The masking mechanism and component parameters are jointly non-identifiable, so we recommend sensitivity analysis over model elaboration. Software is available in the `mdrelax` R package.

Keywords: Series systems, masked failure data, non-informative masking, model misspecification, sensitivity analysis, maximum likelihood estimation

1 Introduction

Estimating the reliability of individual components in a series system is a fundamental challenge when failure causes are masked. When a series system fails, diagnostic procedures may identify only a *candidate set* of components that could have caused the failure, rather than pinpointing the exact failed component [31]. Maximum likelihood estimation under masked data has been developed extensively under a standard set of assumptions [26, 28, 32]: (C1) the true failed component is always in the candidate set; (C2) masking is non-informative, meaning the candidate set formation is independent of the failure cause given the failure time; and (C3) the masking mechanism does not depend on the component reliability parameters.

Each of these conditions is a restriction on the diagnostic process that produces the candidate set. Taken together, they are equivalent to the joint assumption of *coarsening at random* [14, 17] and parameter distinctness [23] for the candidate-set coarsening map. Each may fail in practice. The technician’s experience may make the candidate set informative about the cause (C2 violation). An automated diagnostic may weight candidate inclusion by failure-rate ranking (C3 violation). In the worst case, the diagnostic may miss the failed component entirely, producing a candidate set that does not contain the true cause (C1 violation).

Prior work has addressed each violation in isolation, often in distinct vocabularies. For C2, alternative parametric masking models have been proposed in the reliability literature [5, 11, 18, 21]. Closest in spirit on the reliability side is Jablonka et al. [16], who study estimation for coherent systems with Weibull components under complete and incomplete component-failure information; their relaxation enlarges the observation model whereas we hold the standard C1-C2-C3 model fixed and quantify the bias it incurs when its coarsening conditions are violated. The analogous problem of cause uncertainty in biostatistics has produced parallel methodology: Moreno-Betancur et al. [20] developed pattern-mixture sensitivity analysis for competing risks regression with missing causes; Bakoyannis and Yiannoutsos [1] derived closed-form bias expressions and permissible-range bounds for cumulative incidence under cause misclassification; Bakoyannis et al. [2], in concurrent work,

introduced robustness intervals for Cox cause-specific hazards under binary missing cause. The decision to parameterize an unverifiable departure rather than to model it is the partial-ignorability stance of Harel and Schafer [13], who formalize treating only part of a missingness mechanism as ignorable; the candidate-set coarsening here is exactly such a partially-ignorable mechanism. The local sensitivity of an MLE to assumption violations has been formalized as the Index of Local Sensitivity to Nonignorability (ISNI) [29, 37], with closely related first-order bias expansions in Copas and Eguchi [4] and Siannis et al. [25], and the modern calibrated-bias-bound tradition culminating in the omitted-variable robustness value of Cinelli and Hazlett [3]. The nearest software precedent is the `isni` package [36], which computes ISNI for generalized-linear and mixed models with nonignorable outcome missingness; our `mdrelax` package supplies the set-valued candidate-set reliability instance, which `isni` does not cover.

Three structural features of the masked series-system problem distinguish it from the cause-uncertainty settings treated in the references above. First, the candidate set is *set-valued*, not a single label that may be wrong. The competing-risks treatments cited here represent the cause as fully observed, fully missing, or misclassified to a single alternative label; the set-valued candidate set carries a different likelihood structure, and we show in Appendix C that the two model families are formally non-nested, so set-valued masking does not reduce to point-cause misclassification composed with random coarsening. Second, the standard reliability model carries *three orthogonal coarsening conditions* (C1, C2, C3) whose violations carry distinct semantics, whereas the sensitivity frameworks above are organized around a single scalar departure parameter. Third, the parametric series-system case (Weibull and exponential components) admits closed-form bias expressions; the parallel biostatistics methods we cite are predominantly semiparametric and do not target this parametric instance. Bakoyannis et al. [2] note in their setting that “there is no formula for computing the E-value in the context of competing risks analysis with MNAR causes of failure,” which our closed-form expansions supply for the series-system case.

We address these gaps with a unified sensitivity-analysis framework for masked series-system reliability under calibrated violations of C1, C2, and C3. Our contributions are:

1. **Calibrated severity scales.** We parameterize each coarsening condition on a scalar severity

$\alpha_{C_k} \in [0, \alpha_{C_k}^{\max}]$ with boundary semantics that $\alpha_{C_k} = 0$ recovers the standard C1-C2-C3 model

and $\alpha_{C_k}^{\max}$ marks the transition to an adversarial regime. This aligns with the calibrated-robustness-value philosophy of Cinelli and Hazlett [3] and the pattern-mixture sensitivity parameter of Moreno-Betancur et al. [20], but extends both to a multi-dimensional sensitivity geometry.

2. **Local sensitivity indices for set-valued masking.** We derive closed-form first-order expansions of the misspecified MLE in each α_{C_k} at the standard-model fit, specializing the ISNI construction of Troxel et al. [29] and Zhang and Heitjan [37] from interval-censored time to candidate-set coarsening. The expansions are explicit for exponential components and tractable for Weibull components, addressing the gap that Bakoyannis et al. [2] identify in the cumulative-incidence setting.
3. **Exact, all-orders total-hazard preservation for exponential components.** For exponential components we prove that the standard C1-C2-C3 MLE recovers the true total system hazard $\sum_j \theta_j$ *exactly*, with no bias of any order and regardless of how the candidate sets are formed (Theorem 3.6). The argument is a profile-likelihood factorization that decouples the system-rate score from the candidate-set information entirely. This is strictly stronger than the first-order ISNI guarantee [29] the sensitivity literature provides: the local-sensitivity index for the total hazard is not merely small but identically zero for the constant-hazard family.
4. **Robustness intervals.** For each violation type and each estimand of practical interest (component rates, system hazard, mean time to failure), we construct robustness intervals giving the range of α_{C_k} within which inferential conclusions persist. This extends the framework of Bakoyannis et al. [2] from semiparametric Cox regression with binary missing cause to parametric series systems under set-valued candidate-set coarsening.
5. **Robustness hierarchy across estimands.** We classify which functionals of the parameter vector are protected versus fragile under each violation type. System-level estimands (total hazard, mean time to failure) remain consistently estimated under violations of moderate severity, while component-level estimands degrade in a violation-specific pattern admitting closed-form characterization for exponential systems.
6. **Non-nesting of set-valued and point-cause models.** We prove that the family of joint

cause/candidate-set distributions generated by set-valued Bernoulli masking under relaxed C1 is formally non-nested with the family generated by point-cause misclassification composed with random coarsening (Lemma C.2). This delimits the scope of the framework: it shows that the concurrent point-cause robustness tools of Bakoyannis et al. [2], and the misclassification methods of Moreno-Betancur et al. [20] and Van Rompaye et al. [33], neither subsume nor are subsumed by the set-valued analysis here, so the non-nesting insulates each line of work from the other rather than ranking them.

7. **Software.** The methods are implemented in the `mdrelax` R package, which provides data generation, MLE, Fisher information, sensitivity-index computation, and robustness-interval construction for all model tiers. Existing sensitivity software such as the `isni` package [36] targets generalized-linear and mixed-model outcome missingness; `mdrelax` provides the corresponding tooling for set-valued candidate-set coarsening in series systems, which we are not aware of being available elsewhere, and the closest competing methodology papers [1, 2, 20, 37] do not release accompanying code.

The contributions are linked by the joint coarsening structure of C1, C2, and C3. Together they amount to a sensitivity-analysis toolkit for the practitioner who has fit the standard masked-series-system MLE and wishes to know how robust the resulting reliability estimates are to plausible violations of the modeling assumptions.

We focus throughout on series systems, the $k = 1$ case of the general k -out-of- m family. Companion work [27] treats the complementary identifiability question across general k : there the obstacle is a permutation symmetry of the system likelihood, resolved by a threshold theorem showing that component-level observations of essentially any quality break the symmetry. The present paper instead takes the candidate-set information as given and asks the robustness question: how much do violations of the masking conditions distort the standard C1-C2-C3 estimates?

The remainder of this paper is organized as follows. Section 2 reviews series systems, masked data, the standard C1-C2-C3 likelihood, and the coarsening-at-random framework that subsumes the three conditions. Section 3 develops the unified sensitivity framework: calibrated severity scales, the misspecification theorems and pseudo-true parameters under each violation, local sensitivity indices, and robustness-interval construction. Section 4 presents the simulation sweep across all

three violation types. Section 5 demonstrates the workflow on the turbine engine data of Guo et al. [10]. Section 6 discusses practical implications and limitations, and Section 7 concludes.

2 Background

2.1 Series System Model

Consider a series system composed of m components. The lifetime of the i -th system is

$$T_i = \min\{T_{i1}, T_{i2}, \dots, T_{im}\}, \quad (1)$$

where T_{ij} denotes the lifetime of the j -th component in the i -th system. Component lifetimes are assumed independent with parametric distributions indexed by $\boldsymbol{\theta}_j$; the full parameter vector is $\boldsymbol{\theta} = (\boldsymbol{\theta}_1, \dots, \boldsymbol{\theta}_m) \in \boldsymbol{\Omega}$.

For component j with parameter $\boldsymbol{\theta}_j$, the reliability, density, and hazard functions are

$$R_j(t; \boldsymbol{\theta}_j) = \Pr\{T_{ij} > t\}, \quad f_j(t; \boldsymbol{\theta}_j) = -R'_j(t; \boldsymbol{\theta}_j), \quad h_j(t; \boldsymbol{\theta}_j) = \frac{f_j(t; \boldsymbol{\theta}_j)}{R_j(t; \boldsymbol{\theta}_j)}. \quad (2)$$

Independence yields the series system hazard $h_T(t; \boldsymbol{\theta}) = \sum_{j=1}^m h_j(t; \boldsymbol{\theta}_j)$ and survival $R_T(t; \boldsymbol{\theta}) = \prod_{j=1}^m R_j(t; \boldsymbol{\theta}_j)$.

Given that the system failed at time t , the conditional probability that component j caused the failure is

$$\Pr\{K_i = j \mid T_i = t\} = \frac{h_j(t; \boldsymbol{\theta}_j)}{\sum_{\ell=1}^m h_\ell(t; \boldsymbol{\theta}_\ell)}, \quad (3)$$

which follows from the joint density $f_{T,K}(t, k) = h_k(t; \boldsymbol{\theta}_k) R_T(t; \boldsymbol{\theta})$ and Bayes' theorem [26].

2.2 Masked Data

For each system i , we observe:

- $S_i = \min\{T_i, \tau_i\}$: right-censored system lifetime,
- $\delta_i = \mathbf{1}_{T_i \leq \tau_i}$: event indicator (1 if failure observed),
- $C_i \subseteq \{1, \dots, m\}$: candidate set (only meaningful when $\delta_i = 1$).

The true cause of failure K_i and the individual component lifetimes $(T_{i1}, \dots, T_{im})$ are latent.

2.3 Conditions C1, C2, C3

Tractable likelihood-based inference requires conditions on the relationship between the latent cause K_i and the observed candidate set C_i [10, 31, 32]:

Condition 1 (C1: Failed Component in Candidate Set). $\Pr\{K_i \in C_i\} = 1$.

Condition 2 (C2: Non-Informative Masking). *For all $c \subseteq \{1, \dots, m\}$, all $t > 0$, and all $j, j' \in c$:*

$$\Pr\{C_i = c \mid T_i = t, K_i = j\} = \Pr\{C_i = c \mid T_i = t, K_i = j'\}. \quad (4)$$

Condition 3 (C3: Parameter-Independent Masking). $\Pr\{C_i = c \mid T_i = t, K_i = j\}$ *does not depend on θ .*

C1 is a minimal correctness requirement. C3 is often reasonable when the diagnostic process is fixed independently of the reliability parameters. C2 is the strongest assumption: it requires that the diagnostic process which generates candidate sets reveals nothing about the failure cause beyond C1. This paper studies the consequences of C2 failure.

2.4 Likelihood Under C1-C2-C3

Theorem 2.1 (Likelihood Under C1-C2-C3 [26, 28]). *Under Conditions C1, C2, and C3, the likelihood contribution from observation i is*

$$L_i(\theta) \propto \prod_{\ell=1}^m R_\ell(s_i; \theta_\ell) \cdot \left[\sum_{k \in C_i} h_k(s_i; \theta_k) \right]^{\delta_i}. \quad (5)$$

The masking probability factors out and cancels from the likelihood, leaving a tractable expression for MLE. The complete log-likelihood for n independent systems is

$$\ell(\theta) = \sum_{i=1}^n \left[\sum_{j=1}^m \log R_j(s_i; \theta_j) + \delta_i \log \left(\sum_{k \in C_i} h_k(s_i; \theta_k) \right) \right]. \quad (6)$$

Exponential specialization. When each component has exponential lifetime $T_{ij} \sim \text{Exp}(\theta_j)$ with rate $\theta_j > 0$, we have $h_j(t) = \theta_j$ and $R_j(t) = e^{-\theta_j t}$, giving

$$\ell(\boldsymbol{\theta}) = \sum_{i=1}^n \left[-s_i \sum_{j=1}^m \theta_j + \delta_i \log \left(\sum_{k \in c_i} \theta_k \right) \right]. \quad (7)$$

Weibull specialization. When each component has Weibull lifetime with shape k_j and scale λ_j , the hazard is $h_j(t) = (k_j/\lambda_j)(t/\lambda_j)^{k_j-1}$ and $R_j(t) = \exp[-(t/\lambda_j)^{k_j}]$. The log-likelihood becomes

$$\ell(\boldsymbol{\theta}) = \sum_{i=1}^n \left[-\sum_{j=1}^m \left(\frac{s_i}{\lambda_j} \right)^{k_j} + \delta_i \log \left(\sum_{k \in c_i} h_k(s_i) \right) \right], \quad (8)$$

where $\boldsymbol{\theta} = (k_1, \lambda_1, \dots, k_m, \lambda_m)$.

2.5 Prior Work on Dependent Masking

Several authors have proposed models that relax C2 by explicitly specifying a masking mechanism that depends on the failure cause.

Lin and Guess [18] introduced proportional masking probabilities for two-component systems, deriving closed-form MLEs under the assumption that the masking probability for the true cause exceeds that for the non-failed component by a fixed ratio. Guttman et al. [11] extended two-component dependent masking to a Bayesian framework with conjugate priors. Craiu and Duchesne [6] developed EM-based inference for the competing risks model with masked causes, deriving conditions under which the EM algorithm converges to consistent estimators. Mukhopadhyay and Basu [21] extended Bayesian analysis to general m -component systems where masking probabilities depend on the cause of failure. Craiu and Reiser [5] studied conditional masking probability models and proved identifiability results under specific structural assumptions on the masking mechanism. Flehinger et al. [9] developed parametric Weibull models for masked competing risks with two-stage diagnostic procedures.

These approaches share a common strategy: they replace C2 with a specific parametric masking model and estimate its parameters jointly with the component reliability parameters. The implicit assumption is that the masking model is correctly specified. However, the masking mechanism is itself a diagnostic process that is rarely known in detail.

Our approach is instead inspired by sensitivity analysis methodology in causal inference [22], where the strength of an unverifiable assumption is parameterized and conclusions are examined as the assumption is relaxed. VanderWeele and Ding [34] formalize this via the E-value, quantifying the minimum confounding strength needed to overturn a finding; Siannis [24] apply the same principle to informative censoring in survival analysis. We adapt this paradigm to masked data: rather than proposing an alternative masking model, we ask *how sensitive is the standard C1-C2-C3 estimator to violations of C2?*

3 Sensitivity Framework

We now develop the tools needed to study the sensitivity of C1-C2-C3 inference to C2 violations. We proceed in four steps: the general likelihood under C1 alone (Section 3.1), a Bernoulli model for generating controlled C2 violations (Section 3.2), a misspecification theorem characterizing the resulting bias (Section 3.3), and a non-identifiability result that motivates sensitivity analysis over model estimation (Section 3.5).

3.1 Likelihood Under C1 Alone

When only C1 holds, the masking probability enters the likelihood and cannot be eliminated.

Theorem 3.1 (Likelihood Under C1 Alone). *Under Condition C1 alone, the likelihood contribution from an uncensored observation (s_i, c_i) is:*

$$L_i(\boldsymbol{\theta}) = \prod_{\ell=1}^m R_\ell(s_i; \boldsymbol{\theta}_\ell) \cdot \sum_{k \in c_i} h_k(s_i; \boldsymbol{\theta}_k) \cdot \Pr\{C_i = c_i \mid T_i = s_i, K_i = k\}. \quad (9)$$

Proof. Under C1, $\Pr\{C_i = c \mid T_i = t, K_i = k\} = 0$ when $k \notin c$. Summing over K_i :

$$\begin{aligned} f_{T_i, C_i}(t, c; \boldsymbol{\theta}) &= \sum_{k=1}^m h_k(t; \boldsymbol{\theta}_k) \prod_{\ell=1}^m R_\ell(t; \boldsymbol{\theta}_\ell) \cdot \Pr\{C_i = c \mid T_i = t, K_i = k\} \\ &= \prod_{\ell=1}^m R_\ell(t; \boldsymbol{\theta}_\ell) \cdot \sum_{k \in c} h_k(t; \boldsymbol{\theta}_k) \Pr\{C_i = c \mid T_i = t, K_i = k\}. \quad \square \end{aligned}$$

Under C2, the masking probability $\Pr\{C_i = c \mid T_i = t, K_i = k\}$ is constant over $k \in c$ and factors out of the sum. Under C3, it does not depend on $\boldsymbol{\theta}$ and can be dropped from the likelihood. When

either condition fails, the masking probability remains inside the sum, coupling the inference about θ to the unknown masking mechanism.

3.2 The Bernoulli Perturbation Model

To generate data with controlled C2 violations, we use a Bernoulli candidate set model. We do not claim this model describes how masking works in practice. Rather, it provides a device for producing data whose departure from C2 is known and calibrated.

Definition 3.2 (Bernoulli Masking Model). *Let $p_j(k) = \Pr\{j \in C_i \mid K_i = k\}$ denote the probability that component j is included in the candidate set given that component k failed. Under C1, $p_j(j) = 1$ for all j . These probabilities form an $m \times m$ matrix:*

$$\mathbf{P} = \begin{pmatrix} 1 & p_1(2) & \cdots & p_1(m) \\ p_2(1) & 1 & \cdots & p_2(m) \\ \vdots & \vdots & \ddots & \vdots \\ p_m(1) & p_m(2) & \cdots & 1 \end{pmatrix}, \quad (10)$$

where each component is included independently. The probability of observing candidate set c given $K_i = k$ is

$$\pi_k(c) = \mathbf{1}_{k \in c} \prod_{j \in c \setminus \{k\}} p_j(k) \prod_{j \notin c} (1 - p_j(k)). \quad (11)$$

Remark 3.1 (C2 in Terms of $\mathbf{P}$). Condition C2 holds if and only if each row of $\mathbf{P}$ has constant off-diagonal entries: $p_j(k) = p_j$ for all $k \neq j$. When the columns of $\mathbf{P}$ differ, the masking mechanism “knows” which component failed, violating C2.

The likelihood under this model follows immediately from Theorem 3.1:

Corollary 3.3 (Likelihood Under the Bernoulli Model). *Under the Bernoulli masking model with known $\mathbf{P}$, the likelihood contribution from an uncensored observation (s_i, c_i) is:*

$$L_i(\theta) = \prod_{\ell=1}^m R_\ell(s_i; \theta_\ell) \cdot \sum_{k \in c_i} h_k(s_i; \theta_k) \cdot \pi_k(c_i). \quad (12)$$

Parameterizing violation severity. We generate a family of $\mathbf{P}$ matrices indexed by a scalar severity parameter $s \in [0, 1]$:

$$\mathbf{P}(s) = \mathbf{P}_0 + s \mathbf{D}, \quad (13)$$

where $\mathbf{P}_0$ is the uniform (C2-satisfying) matrix with all off-diagonal entries equal to a base probability p_0 , and $\mathbf{D}$ is a fixed “direction” matrix with zero diagonal, whose off-diagonal entries create column-wise asymmetry. Off-diagonal entries of $\mathbf{P}(s)$ are clamped to $[0.05, 0.95]$ to maintain valid probabilities. At $s = 0$, C2 holds exactly; as s increases, the columns of $\mathbf{P}$ diverge and the masking becomes increasingly informative.

3.3 Misspecification Under C2 Violation

We now characterize what happens when the standard C1-C2-C3 model is fitted to data generated under C2 violation.

Theorem 3.4 (Pseudo-True Parameter Under C2 Misspecification). *Suppose data is generated under C1 and C3 with informative masking weights $\pi_{k,c}$, but estimation is performed under C1-C2-C3 (assuming non-informative masking). By the general theory of misspecified MLEs [15, 35], the MLE under the misspecified model converges in probability to a pseudo-true parameter $\boldsymbol{\theta}^\dagger$ satisfying*

$$\mathbb{E}_{\boldsymbol{\theta}^*} \left[\frac{\partial \ell^{\text{wrong}}}{\partial \theta_j} \Big|_{\boldsymbol{\theta}=\boldsymbol{\theta}^\dagger} \right] = 0, \quad (14)$$

where the expectation is taken under the true data-generating process with parameter $\boldsymbol{\theta}^*$. For exponential components with rates $\boldsymbol{\theta} = (\theta_1, \dots, \theta_m)$, the pseudo-true parameter differs from $\boldsymbol{\theta}^*$ whenever the masking weights $\pi_{k,c}$ vary with k , i.e., whenever C2 is violated.

Proof. The misspecified score for exponential component j is

$$\frac{\partial \ell_i^{\text{wrong}}}{\partial \theta_j} = -s_i + \delta_i \frac{\mathbf{1}_{j \in c_i}}{\sum_{k \in c_i} \theta_k}. \quad (15)$$

The true score under C1-C3 (with masking weights $\pi_{k,c}$) is

$$\frac{\partial \ell_i^{\text{true}}}{\partial \theta_j} = -s_i + \delta_i \frac{\pi_{j,c_i} \mathbf{1}_{j \in c_i}}{\sum_{k \in c_i} \theta_k \pi_{k,c_i}}. \quad (16)$$

At θ^* , the true score has expectation zero. The misspecified score replaces the weights $\pi_{k,c}$ with uniform weights. Setting $\mathbb{E}[\partial \ell^{\text{wrong}} / \partial \theta_j] = 0$ at $\theta^\dagger$ yields a system whose solution differs from θ^* whenever the masking weights vary with k . $\square$

Proposition 3.5 (System Hazard Robustness, exponential components). *Let the components be exponential, so that the system hazard is $S = \sum_j \theta_j$, and suppose the retained system lifetimes satisfy $T \sim \text{Exp}(S^*)$ with $S^* = \sum_j \theta_j^*$ (the masking mechanism produces retention events independent of T ; see Theorem 3.6 and its corollaries). Then the misspecified C1-C2-C3 MLE preserves the total system hazard exactly, $\sum_j \theta_j^\dagger = S^*$, for any severity of C2 violation.*

Proof. This is the specialization of Theorem 3.6 to the C2 violation model, proved there via the profile-likelihood factorization of the exponential log-likelihood into an S -only and a ϕ -only part; Corollary 3.7 states the C2 case explicitly. The estimand $\sum_j \theta_j$ is the system hazard precisely because the components are exponential. $\square$

Remark 3.2 (Interpretation and scope). The misspecified estimator acts as if each candidate in c_i contributes equally to the observed failure, regardless of the actual masking weights. When a component is over-included in candidate sets (its column of $\mathbf{P}$ has higher off-diagonal values), it “claims credit” for more failures than it actually caused, inflating its estimated rate. The total system hazard is protected because, for exponential components, the system-hazard MLE $\hat{S} = n_F / \sum_i s_i$ depends on the data only through the failure count and total exposure, neither of which is altered by how failures are attributed across candidates. For non-exponential components $\sum_j \theta_j$ is no longer the system hazard and the exact factorization fails; total-hazard robustness then weakens to first-order preservation in the violation severity, which we report empirically rather than claim as an exact result.

3.4 Exact Total-Hazard Preservation for Exponential Components

The total-hazard robustness of Proposition 3.5 holds approximately for general parametric components and is strengthened to an exact result for the exponential case via a profile-likelihood argument. The result applies not only to C2 misspecification but to any masking violation, including C1 and C3 violations to be analyzed in Sections 3.6 and 3.7.

Theorem 3.6 (Exact Total-Hazard Preservation, Exponential Components). *For an m -component series system with exponential components and true rates $\boldsymbol{\theta}^* = (\theta_1^*, \dots, \theta_m^*)$, suppose data are generated under any masking model such that retained system lifetimes satisfy $T \sim \text{Exp}(S^*)$, where $S^* = \sum_j \theta_j^*$. The standard C1-C2-C3 MLE applied to retained observations then satisfies*

$$\sum_{j=1}^m \theta_j^\dagger = S^* \quad (17)$$

exactly, with no bias of any order in the total system hazard, regardless of the severity of the masking violation.

Proof. Reparameterize $\theta_j = S\phi_j$ with $S = \sum_j \theta_j$ and ϕ lying on the open simplex. The standard C1-C2-C3 log-likelihood for exponential components is

$$\ell^{\text{wrong}}(\boldsymbol{\theta}) = -S \sum_{i=1}^n s_i + \sum_{i:\delta_i=1} \log \left(\sum_{k \in c_i} \theta_k \right), \quad (18)$$

and substituting $\theta_k = S\phi_k$:

$$\ell^{\text{wrong}}(S, \phi) = \underbrace{-S \sum_i s_i + n_F \log S}_{\ell_S(S)} + \underbrace{\sum_{i:\delta_i=1} \log W_{c_i}(\phi)}_{\ell_\phi(\phi)}, \quad (19)$$

where $n_F = |\{i : \delta_i = 1\}|$ and $W_c(\phi) = \sum_{k \in c} \phi_k$. The log-likelihood is the sum of an S -only part and a ϕ -only part. The score for S is therefore $\partial \ell^{\text{wrong}} / \partial S = -\sum_i s_i + n_F / S$, independent of ϕ , yielding the MLE

$$\hat{S} = \frac{n_F}{\sum_i s_i}. \quad (20)$$

This is the standard exponential rate MLE for type-I censored data with n_F observed failures and total observation time $\sum_i s_i$. Under the assumed retention distribution $T \sim \text{Exp}(S^*)$, $\hat{S} \rightarrow S^*$ in probability, hence $\sum_j \theta_j^\dagger = S^*$. $\square$

Remark 3.3 (Scope). The result rests on two structural features. First, the exponential log-likelihood factors additively as in (19), a consequence of the exponential hazard being constant in time. For Weibull or other non-constant-hazard parametric components, no such factorization holds, and

total-hazard robustness generally weakens to first-order preservation in the violation severity. Second, the marginal distribution of retained system lifetimes must be $\text{Exp}(S^*)$, which holds whenever the masking mechanism produces retention events independent of T . We verify this condition for the Bernoulli C1, C2, and C3 violation models considered in this paper as corollaries below.

Corollary 3.7 (Strengthening of Proposition 3.5 for exponential). *For exponential components under the Bernoulli C2 violation model with C1 holding, the misspecified C1-C2-C3 MLE preserves S^* exactly, not just approximately, for any severity of C2 violation.*

Proof. Under any Bernoulli C2 violation model with C1 holding, $|C| \geq 1$ almost surely and no retention is needed. The marginal distribution of T is $\text{Exp}(S^*)$, so Theorem 3.6 applies. $\square$

3.5 The Identifiability Trap

A natural response to C2 uncertainty would be to estimate $\mathbf{P}$ jointly with $\boldsymbol{\theta}$ from the data. This fails.

Theorem 3.8 (Non-Identifiability of $(\boldsymbol{\theta}, \mathbf{P})$). *For exponential series systems with masked data under the Bernoulli masking model, the joint parameter $(\boldsymbol{\theta}, \mathbf{P})$ is not identifiable from the observed data $(s_i, c_i, \delta_i)_{i=1}^n$. Specifically, for any $(\boldsymbol{\theta}^*, \mathbf{P}^*)$ with $\boldsymbol{\theta}^* \neq \boldsymbol{\theta}'$ there exists a one-parameter family of $(\boldsymbol{\theta}', \mathbf{P}')$ yielding the same likelihood. However, the total system hazard $\sum_{j=1}^m \theta_j$ remains identifiable from the marginal system lifetime data.*

Proof. We prove the result constructively for $m = 2$ components; the general case follows analogously. The observed-data likelihood under the Bernoulli model depends on $(\theta_1, \theta_2, p_{12}, p_{21})$ through three functionals: (i) the system hazard $S = \theta_1 + \theta_2$, which determines the system survival $R_T(t) = e^{-St}$; (ii) the weighted hazard contribution for candidate set $c = \{1, 2\}$: $\theta_1 p_{21} + \theta_2 p_{12}$; and (iii) the single-component hazard contribution: $\theta_1(1 - p_{21})$ for $c = \{1\}$, and $\theta_2(1 - p_{12})$ for $c = \{2\}$.

This gives four unknowns $(\theta_1, \theta_2, p_{12}, p_{21})$ constrained by three likelihood-determining quantities, leaving a one-dimensional non-identifiable family. Explicitly: given $(\theta_1^*, \theta_2^*, p_{12}^*, p_{21}^*)$, define for any ε

with $|\varepsilon|$ small enough that all parameters remain in their domains:

$$\theta'_1 = \theta_1^* + \varepsilon, \quad \theta'_2 = \theta_2^* - \varepsilon, \quad (21)$$

$$p'_{21} = 1 - \frac{\theta_1^*(1 - p_{21}^*)}{\theta_1^* + \varepsilon}, \quad p'_{12} = 1 - \frac{\theta_2^*(1 - p_{12}^*)}{\theta_2^* - \varepsilon}. \quad (22)$$

Then $S' = S^*$, and direct substitution verifies that all three functionals are preserved, so $(\boldsymbol{\theta}', \mathbf{P}')$ yields the same likelihood as $(\boldsymbol{\theta}^*, \mathbf{P}^*)$.

Total hazard identifiability follows from the marginal system lifetime: $R_T(t) = e^{-S_t}$ depends only on S , which is identifiable from the (uncensored or right-censored) system failure times alone, without reference to the candidate sets. $\square$

This extends the classical non-identifiability result for competing risks [7, 30] to the masked data setting: the masking mechanism and component parameters play analogous roles to the dependence structure and marginal distributions in the competing risks framework [19].

This result is supported by both theoretical argument and simulation evidence. Even with $n = 2000$ observations, the joint MLE of $(\boldsymbol{\theta}, \mathbf{P})$ exhibits persistent bias with individual component rates confounded with the off-diagonal entries of $\mathbf{P}$, while the total hazard $\sum_j \hat{\theta}_j$ converges to the true value (see Appendix B).

Remark 3.4 (Implications). Since $\boldsymbol{\theta}$ and $\mathbf{P}$ cannot be disentangled from the data, we cannot “just estimate” the masking mechanism. This makes sensitivity analysis the appropriate tool: fit the standard C1-C2-C3 model, then assess how conclusions change under plausible perturbations of the masking structure via different $\mathbf{P}$ matrices. If the substantive conclusions are stable, C2 violations are not a concern for the application at hand.

3.6 Sensitivity to C1 Violations

We now extend the framework to violations of C1, the assumption that the failed component is in the candidate set. The construction is parallel to the C2 case of Sections 3.2 and 3.3: we introduce a calibrated scalar severity parameter, derive the misspecified score equation, and expand the pseudo-true parameter to first order.

The Bernoulli C1 violation model. Within the Bernoulli masking model of Definition 3.2, a C1 violation is induced by setting

$$p_k(k) = 1 - \alpha_{C_1}, \quad p_j(k) = p_0 \text{ for } j \neq k. \quad (23)$$

The off-diagonal probabilities are constant in k , so C2 continues to hold along the C1 sweep, isolating the effect of C1 violation. The parameter $\alpha_{C_1} \in [0, 1/2]$ measures the probability that the true cause is excluded from the candidate set, with $\alpha_{C_1} = 0$ recovering the standard model and $\alpha_{C_1} = 1/2$ marking the boundary beyond which the candidate set is anti-informative (the true cause is more likely excluded than included).

Under (23) the candidate set may be empty with probability $\alpha_{C_1}(1 - p_0)^{m-1}$ conditional on a failure. The standard C1-C2-C3 likelihood is undefined on empty candidate sets, so we condition on $|C| \geq 1$ throughout the analysis. This matches the practitioner workflow: failures with no diagnostic information are excluded from the parametric MLE.

Theorem 3.9 (Pseudo-True Parameter Under C1 Misspecification, $m = 2$). *For a two-component exponential series system with rates $\boldsymbol{\theta}^* = (\theta_1^*, \theta_2^*)$ and data generated under the Bernoulli C1 violation model (23), conditioning on $|C| \geq 1$, the misspecified C1-C2-C3 MLE converges to a pseudo-true parameter $\boldsymbol{\theta}^\dagger(\alpha_{C_1})$ admitting the first-order expansion*

$$\theta_j^\dagger = \theta_j^* + \alpha_{C_1} \Delta_j^{C_1}(\boldsymbol{\theta}^*; p_0) + O(\alpha_{C_1}^2), \quad (24)$$

with

$$\Delta_1^{C_1} = \frac{p_0(\theta_2^* - \theta_1^*)}{1 - p_0}, \quad \Delta_2^{C_1} = -\Delta_1^{C_1} = \frac{p_0(\theta_1^* - \theta_2^*)}{1 - p_0}. \quad (25)$$

Proof. The misspecified score for exponential component j is $\partial \ell_i^{\text{wrong}} / \partial \theta_j = -s_i + \delta_i \mathbf{1}_{j \in c_i} / \sum_{k \in c_i} \theta_k$. Setting $\mathbb{E}_{\boldsymbol{\theta}^*}[\partial \ell^{\text{wrong}} / \partial \theta_j] = 0$ at $\boldsymbol{\theta} = \boldsymbol{\theta}^\dagger$, and computing the expectation under the Bernoulli C1 violation model with $|C| \geq 1$ and no censoring, the score equation for $\theta_1^\dagger$ reduces to (after dividing by S^* and multiplying by the conditional-probability normalizer $Z = 1 - \alpha_{C_1}(1 - p_0)$):

$$Z = \frac{\theta_1^*(1 - \alpha_{C_1})(1 - p_0) + \theta_2^* \alpha_{C_1} p_0}{\theta_1^\dagger} + \frac{p_0(1 - \alpha_{C_1})S^*}{\theta_1^\dagger + \theta_2^\dagger}, \quad (26)$$

with the analogous equation for $\theta_2^\dagger$ obtained by swapping indices. Writing $\theta_j^\dagger = \theta_j^* + \alpha_{C_1} \Delta_j + O(\alpha_{C_1}^2)$ and expanding both sides of (26) to first order in α_{C_1} , the zeroth-order terms vanish identically. The first-order coefficients yield the linear system

$$(1 - p_0) \frac{\Delta_1}{\theta_1^*} + p_0 \frac{\Delta_1 + \Delta_2}{S^*} = p_0 \frac{\theta_2^* - \theta_1^*}{\theta_1^*}, \quad (27)$$

$$(1 - p_0) \frac{\Delta_2}{\theta_2^*} + p_0 \frac{\Delta_1 + \Delta_2}{S^*} = p_0 \frac{\theta_1^* - \theta_2^*}{\theta_2^*}. \quad (28)$$

Setting $a = \Delta_1/\theta_1^*$, $b = \Delta_2/\theta_2^*$, and $r = \theta_2^*/\theta_1^*$, the system (27), (28) reduces to two linear equations in (a, b) . Combining them yields $a + br = 0$, i.e., $\Delta_1 + \Delta_2 = 0$. Substituting back gives $a = p_0(r - 1)/(1 - p_0)$, hence $\Delta_1 = \theta_1^* \cdot p_0(r - 1)/(1 - p_0) = p_0(\theta_2^* - \theta_1^*)/(1 - p_0)$, and similarly $\Delta_2 = -\Delta_1$. $\square$

Corollary 3.10 (Total Hazard Preservation Under C1 Violation, Exponential, General m). *For exponential series systems of any size $m \geq 2$ under the Bernoulli C1 violation model (23) with retention on $|C| \geq 1$, the misspecified MLE preserves the total system hazard exactly: $\sum_j \theta_j^\dagger = S^*$ for all $\alpha_{C_1} \in [0, 1/2]$. In particular, the $m = 2$ first-order coefficients in (25) satisfy $\Delta_1^{C_1} + \Delta_2^{C_1} = 0$, and this preservation holds at all orders for general m .*

Proof. By Theorem 3.6, it suffices to verify that retained system lifetimes have the marginal distribution $\text{Exp}(S^*)$. Under (23), $\Pr\{|C| \geq 1 \mid K = k, T = t\} = 1 - \alpha_{C_1}(1 - p_0)^{m-1}$ is independent of both k and t . The retention event $\{|C| \geq 1\}$ is therefore independent of T , and the conditional distribution $T \mid |C| \geq 1$ equals the marginal $T \sim \text{Exp}(S^*)$. $\square$

Remark 3.5 (Interpretation). When the candidate set excludes the true cause, the misspecified MLE attributes the failure to whichever non-failed component appears in the candidate set. Equation (25) shows that the first-order bias is proportional to the rate gap $\theta_2^* - \theta_1^*$: the higher-rate component absorbs failures it did not cause from the lower-rate component, biasing its estimate upward by the same magnitude that the lower-rate component is biased downward. The total system hazard is preserved because the system lifetime $T \sim \text{Exp}(S^*)$ identifies S^* directly from time data, independent of how individual failures are attributed. The amplification factor $p_0/(1 - p_0)$ reflects how much the surviving candidate set covers non-failed components; it diverges as $p_0 \rightarrow 1$, where individual rates become non-identifiable even under C1.

The proof generalizes to $m \geq 3$ components with the same total-hazard preservation result. The structure of the linear system (score Jacobian on the left, α_{C_k} -derivative of the score on the right) is identical to (31) and is implemented numerically for any m in the `mdrelax` R package (Appendix D).

3.7 Sensitivity to C3 Violations

C3 parameterizes whether the masking probability depends on the component reliability parameters θ . We adopt the power-weight model [21]:

$$p_j(k; \theta) = p_0 \cdot \frac{\theta_j^{\alpha_{C_3}}}{\max_l \theta_l^{\alpha_{C_3}}} \quad \text{for } j \neq k, \quad (29)$$

with $p_k(k) = 1$ (C1 holds). The normalizer $\max_l \theta_l^{\alpha_{C_3}}$ is independent of the failed component k , so the inclusion weight of component j is the same regardless of which component failed: C2 holds and the violation is purely C3 (dependence of the masking probability on θ). This is the model implemented in `wei_inclusion_probs` (`R/wei_series_c1-c2.R`), where the weights are normalized by their global maximum. The scalar $\alpha_{C_3} \in [0, \alpha_{C_3}^{\max}]$ is the calibrated severity: $\alpha_{C_3} = 0$ recovers constant masking $p_j(k) = p_0$ (C3 holds), while increasing α_{C_3} makes high-rate components more likely to appear in candidate sets, modeling a diagnostic procedure that weights candidates by their estimated reliability ranking.

Theorem 3.11 (Pseudo-True Parameter Under C3 Misspecification). *For exponential series systems with data generated under the power-weight C3 violation (29), the misspecified C1-C2-C3 MLE converges to a pseudo-true parameter $\theta^\dagger(\alpha_{C_3})$ admitting the first-order expansion*

$$\theta_j^\dagger = \theta_j^* + \alpha_{C_3} \Delta_j^{C_3}(\theta^*; p_0) + O(\alpha_{C_3}^2). \quad (30)$$

The first-order coefficients satisfy

1. **Total preservation, general m :** $\sum_{j=1}^m \Delta_j^{C_3}(\theta^*; p_0) = 0$, and the all-orders identity $\sum_j \theta_j^\dagger(\alpha_{C_3}) = S^*$ holds for every α_{C_3} (by Theorem 3.6, since the power-weight model makes $|C| \geq 1$ deterministic so the retention condition is satisfied trivially).
2. **Linear system for general m :** the vector $\Delta^{C_3}(\theta^*; p_0) \in \mathbb{R}^m$ is the unique solution (on the

rate simplex) of the score-Jacobian system

$$\mathbf{J}(\boldsymbol{\theta}^*) \Delta^{C_3}(\boldsymbol{\theta}^*; p_0) = \mathbf{r}(\boldsymbol{\theta}^*; p_0), \quad (31)$$

where $\mathbf{J}(\boldsymbol{\theta}^*)$ is the Jacobian of the misspecified score in $\boldsymbol{\theta}$ at $\boldsymbol{\theta}^*$ (the standard expected information matrix evaluated under $\alpha_{C_3} = 0$) and $\mathbf{r}(\boldsymbol{\theta}^*; p_0)$ is the partial derivative of the misspecified score in α_{C_3} at $\alpha_{C_3} = 0$ evaluated under the power-weight perturbation. Both matrices are rational functions of $(\boldsymbol{\theta}^*, p_0)$ obtained by differentiating the misspecified score in Theorem 3.4. The total-preservation identity in item 1 forces $\mathbf{1}^\top \Delta^{C_3} = 0$, so the system has rank $m - 1$ on the simplex; the rank deficiency is exactly the directional ambiguity that the masking-mechanism non-identifiability of Theorem 3.8 predicts.

The simulation results in Section 4 demonstrate the qualitative behavior of $\Delta_j^{C_3}$: rates of high-failure-rate components are inflated further (positive bias amplification), while low-rate components are underestimated. The component-level redistribution respects the total-hazard constraint for exponential components. For Weibull components, where the factorization (19) does not apply, total-hazard preservation under C3 is no longer automatic and is studied empirically in Section 4.

3.8 Local Sensitivity Indices and the Robustness Hierarchy

The first-order coefficients $\Delta_j^{C_k}$ from Theorem 3.4, Theorem 3.9, and Theorem 3.11 specialize the local-sensitivity construction of Troxel et al. [29] and Zhang and Heitjan [37] to candidate-set coarsening. We collect them into a single index and use it to classify estimands by robustness.

Definition 3.12 (Local Sensitivity Index for Candidate-Set Coarsening). *For a functional $\psi(\boldsymbol{\theta})$ of the parameter vector and a violation type $C_k \in \{C_1, C_2, C_3\}$, the local sensitivity index is*

$$\text{ISNI}_{C_k}[\psi] := \left. \frac{\partial}{\partial \alpha_{C_k}} \psi(\boldsymbol{\theta}^\dagger(\alpha_{C_k})) \right|_{\alpha_{C_k}=0} = \nabla \psi(\boldsymbol{\theta}^*)^\top \Delta^{C_k}(\boldsymbol{\theta}^*). \quad (32)$$

Robustness intervals. The local sensitivity index converts an unverifiable masking deviation α_{C_k} into a bound on the asymptotic bias of ψ under the standard MLE. We package that bound as an interval covering the pseudo-true value $\psi(\boldsymbol{\theta}^\dagger(\alpha_{C_k}))$ across a practitioner-chosen plausibility set for α_{C_k} .

Definition 3.13 (First-Order Robustness Interval). *Fix a functional $\psi: \Theta \rightarrow \mathbb{R}$ that is C^2 in a neighborhood of the standard-model fit θ^* , a violation type C_k , and a plausibility set $\mathcal{A}_k \subset [0, \alpha_{C_k}^{\max}]$. The first-order robustness interval for ψ under C_k is*

$$\text{RI}_{C_k}^{(1)}(\psi; \mathcal{A}_k) := \left[\hat{\psi} - |\text{ISNI}_{C_k}[\psi]| \sup_{\alpha \in \mathcal{A}_k} \alpha, \hat{\psi} + |\text{ISNI}_{C_k}[\psi]| \sup_{\alpha \in \mathcal{A}_k} \alpha \right], \quad (33)$$

where $\hat{\psi} = \psi(\hat{\theta})$ is the plug-in estimate from the standard MLE.

Proposition 3.14 (Asymptotic Coverage of $\psi(\theta^\dagger)$). *Assume the regularity conditions of Theorems 3.4, 3.9 and 3.11: the standard-model fit $\hat{\theta}$ is $\sqrt{n}$ -consistent for θ^* when $\alpha_{C_k} = 0$; the pseudo-true map $\alpha_{C_k} \mapsto \theta^\dagger(\alpha_{C_k})$ is C^2 on a neighborhood of 0 with a uniform bound M_k on the operator norm of its Hessian and a uniform bound D_k on the norm of its first derivative $\dot{\theta} = d\theta^\dagger/d\alpha_{C_k}$ on $\mathcal{A}_k$; and ψ is C^2 with $\|\nabla\psi\|$ bounded by L_ψ and $\|\nabla^2\psi\|$ bounded by B_ψ on a neighborhood of θ^* . Then for every $\alpha_{C_k} \in \mathcal{A}_k$ the pseudo-true value admits the expansion*

$$\psi(\theta^\dagger(\alpha_{C_k})) = \psi(\theta^*) + \alpha_{C_k} \text{ISNI}_{C_k}[\psi] + R_k(\alpha_{C_k}), \quad |R_k(\alpha_{C_k})| \leq \frac{1}{2} (L_\psi M_k + B_\psi D_k^2) \alpha_{C_k}^2, \quad (34)$$

with $L_\psi = \sup_{\theta \in \mathcal{N}(\theta^*)} \|\nabla\psi(\theta)\|$ and $B_\psi = \sup_{\theta \in \mathcal{N}(\theta^*)} \|\nabla^2\psi(\theta)\|$. Consequently, $\text{RI}_{C_k}^{(1)}(\psi; \mathcal{A}_k)$ contains $\psi(\theta^\dagger(\alpha_{C_k}))$ for every $\alpha_{C_k} \in \mathcal{A}_k$ up to a remainder that is $O(\sup \mathcal{A}_k^2)$.

Proof. Write $f(\alpha) = \psi(\theta^\dagger(\alpha))$. By the chain rule, $f'(\alpha) = \nabla\psi(\theta^\dagger)^\top \dot{\theta}$ and

$$f''(\alpha) = \dot{\theta}^\top \nabla^2\psi(\theta^\dagger) \dot{\theta} + \nabla\psi(\theta^\dagger)^\top \ddot{\theta},$$

so the second-order term carries *both* the curvature of the pseudo-true map ($\ddot{\theta}$, bounded by M_k) and the curvature of the estimand ($\nabla^2\psi$ contracted with $\dot{\theta}^{\otimes 2}$, bounded by $B_\psi D_k^2$). The linear term is $f'(0) = \text{ISNI}_{C_k}[\psi]$. The Lagrange remainder of the first-order Taylor expansion is $\frac{1}{2}f''(\xi)\alpha^2$ for some ξ , and $|f''(\xi)| \leq L_\psi M_k + B_\psi D_k^2$ uniformly on $\mathcal{A}_k$, giving (34). The estimand-curvature term $B_\psi D_k^2$ is nonzero for every nonlinear ψ (MTTF, reliability, hazard-at- t) and vanishes only for affine ψ . The interval in (33) contains the linear term for every $\alpha_{C_k} \in \mathcal{A}_k$ by construction. $\square$

Remark 3.6 (How to use the interval). The practitioner picks $\mathcal{A}_k = [0, \alpha_{C_k}^*]$ where $\alpha_{C_k}^*$ is the largest violation deemed plausible for the diagnostic process at hand: the default scale is calibrated so that

$\alpha_{C_k}^* = \alpha_{C_k}^{\max}$ is the boundary of the adversarial regime. The ISNI value $\text{ISNI}_{C_k}[\psi]$ is obtained in closed form from $\nabla\psi(\hat{\boldsymbol{\theta}})$ and the $\Delta^{C_k}(\hat{\boldsymbol{\theta}})$ expression from Theorems 3.4, 3.9 and 3.11. The interval is deliberately conservative: it bounds the asymptotic pseudo-true bias and does not absorb Monte Carlo uncertainty in $\hat{\boldsymbol{\theta}}$, which can be added by widening the interval by the usual Wald half-width $1.96 \widehat{\text{se}}(\hat{\psi})$.

Remark 3.7 (When the first-order interval collapses). For estimands that the hierarchy classifies as exactly preserved (items 1 and 2 of Corollary 3.15) the ISNI is zero, $\text{RI}_{C_k}^{(1)}$ collapses to the point estimate, and Proposition 3.14 becomes the assertion that the pseudo-true value coincides with the standard-model fit at every order. For estimands that are first-order robust but not exactly preserved (item 3) the interval has zero linear half-width and only the second-order remainder R_k matters; in practice this remainder is absorbed into the Monte Carlo or sandwich variance. For estimands that are not first-order robust (item 4) the interval width scales linearly with $\sup \mathcal{A}_k$, and is the operative bound for practitioner-facing component-level claims.

Corollary 3.15 (Robustness Hierarchy). *For masked series systems under the standard C1-C2-C3 MLE:*

1. **Exponential components, total system hazard** $S(\boldsymbol{\theta}) = \sum_j \theta_j$: *preserved exactly under any violation of C1, C2, or C3 (Theorem 3.6, Corollary 3.7, Corollary 3.10), giving $\text{ISNI}_{C_k}[S] = 0$ for all k .*
2. **Exponential components, mean time to system failure** $1/S(\boldsymbol{\theta})$: *preserved exactly under any of the three violations (immediate from item 1).*
3. **Weibull components, system-level estimands (system hazard at fixed t , MTTF)**: *first-order robust under C2 and C3 violations ($\text{ISNI}_{C_2}[S] = \text{ISNI}_{C_3}[S] = 0$ at the standard-model fit). Under C1, the first-order coefficient is nonzero but the absolute bias remains bounded across the practical violation range $\alpha_{C_1} \in [0, 0.5]$, with empirical magnitude below 4% in the 5-component sweep of Section 4.*
4. **Individual component parameters** (rates θ_j for exponential, shape-scale pairs for Weibull): *not first-order robust under any of the three violations. Scale parameters of low-rate components exhibit the most catastrophic degradation.*

The asymmetry in item 3 has a structural origin. The exact-preservation proof for exponential (Theorem 3.6) relies on the log-likelihood factorization $\ell(S, \phi) = \ell_S(S) + \ell_\phi(\phi)$, which decouples system-level inference from component-level inference. For Weibull components the analogous factorization fails: the time-portion of the likelihood involves the individual shape-scale pairs that also appear in the candidate-set portion. Under C2 and C3 violations, the candidate-set bias remains balanced enough that it does not propagate to the time portion at first order. Under C1 violation, the candidate set sometimes omits the true cause entirely, producing more aggressive candidate-set bias that does propagate to the time-portion estimates at first order. The bias remains bounded because the system lifetime $T \sim \text{Weibull}_m^{\text{series}}(\theta^*)$ still satisfies the marginal-distribution preservation that powers the exponential argument; only the parametric mapping back to $S(t)$ is biased.

The hierarchy formalizes the empirical pattern observed in Section 4. The practical takeaway is sharp: practitioners reporting system-level reliability estimates from exponential models can ignore concerns about masking-condition violations entirely. For Weibull-component systems, system-level estimates are first-order protected against C2 and C3 violations and admit a small, bounded bias under C1. Component-level estimates are sensitive to all three violations regardless of the component family, and require sensitivity analysis before being reported. We discuss the practical consequences in Section 6.

4 Simulation Study

We conduct a systematic sweep of all three coarsening-condition violations (C1, C2, C3) for a 5-component Weibull series system, fitting the standard C1-C2-C3 model throughout. The empirical results validate the robustness hierarchy of Corollary 3.15 at two sample sizes and quantify the practical impact of each violation on system-level and component-level inference.

4.1 Design

System configuration. We consider an $m = 5$ component Weibull series system with shapes $\mathbf{k}^* = (2.0, 1.5, 1.2, 1.8, 1.0)$ and scales $\boldsymbol{\lambda}^* = (3.0, 4.0, 5.0, 3.5, 4.5)$. The fifth component ($k_5 = 1.0$) is exponential, providing a within-system robustness baseline. Right-censoring time is $\tau = 5$, sample size $n = 500$ (main), and $B = 200$ replications per severity level. The true system hazard at reference

time $t_0 = 2.0$ is $h_T^*(2.0) = 1.460$.

Sweep parameterizations. The three violations are swept independently along their respective calibrated severity scales:

- **C1 sweep:** $\alpha_{C_1} \in \{0, 0.05, 0.10, \dots, 0.50\}$ (11 levels). The Bernoulli model is parameterized as $p_k(k) = 1 - \alpha_{C_1}$ and $p_j(k) = p_0$ for $j \neq k$ with $p_0 = 0.5$ (see (23)). Failures with empty candidate sets are dropped to match the practitioner workflow.
- **C2 sweep:** severity $s \in \{0, 0.1, \dots, 1.0\}$ (11 levels). The P-matrix follows $\mathbf{P}(s) = \mathbf{P}_0 + s \mathbf{D}$ per (13), with $p_0 = 0.5$ and direction matrix $\mathbf{D}$ having a cyclic structure with rows summing to zero (the explicit form is the matrix `make_sweep_P` in `paper/run_sensitivity_sweep.R`).
- **C3 sweep:** $\alpha_{C_3} \in \{0, 0.25, 0.50, \dots, 2.0\}$ (9 levels). The power-weight model from (29) with $p_0 = 0.5$.

Estimation. At each severity level, all B datasets are fitted with the standard C1-C2-C3 Weibull series MLE, ignoring any masking violation. Optimization uses L-BFGS-B with starting values at 90% of true parameters. Across all three sweeps and 2,000+ datasets at the main sample size, every fit converged.

Metrics. For each parameter and severity level, we report bias, RMSE, and 95% Wald coverage. The system hazard $\hat{h}_T(t_0)$ at $t_0 = 2.0$ is tracked separately as the system-level summary, with delta-method standard errors propagated from the parametric MLE.

4.2 Headline: System-Level Robustness

Figure 1 shows the system-hazard bias trajectory for all three violations. The bias stays within $\pm 6\%$ across the entire severity range for every violation (the worst single point being the C1 trajectory, which stays under 4% throughout; see the per-violation maxima below), validating the robustness hierarchy of Corollary 3.15 item 3.

Concretely, at the maximum severity considered for each violation, the system-hazard relative bias is -3.2% (C1, $\alpha_{C_1} = 0.5$), $+2.6\%$ (C2, $s = 1.0$), and -0.4% (C3, $\alpha_{C_3} = 2.0$). For C2 and C3

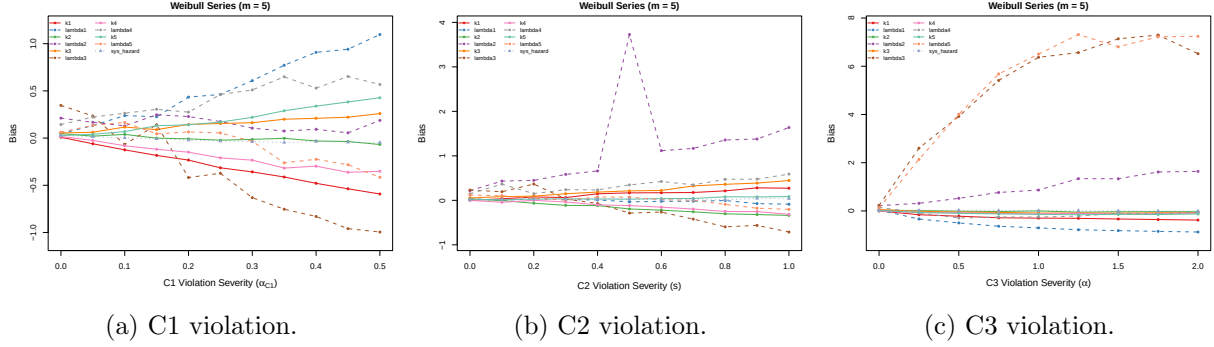


Figure 1: Bias as a function of violation severity for each of C1, C2, C3. The system hazard at $t_0 = 2.0$ (triangle markers) remains bounded within a few percent of the truth across the full severity range under all three violations, while individual component parameters degrade.

the bias is within a few percentage points of finite-sample Monte Carlo noise; for C1 the bias is real but bounded under 4% throughout. A reliability engineer reporting system-level estimates (total hazard, MTTF, system survival at a fixed time) from the standard C1-C2-C3 MLE has substantial protection against all three masking-condition violations, even at severe levels.

4.3 Contrast: Component-Level Fragility

Individual component parameter estimates degrade with severity for all three violations, and the degradation is catastrophic for certain parameter classes. Table 1 summarizes the worst relative biases observed across components for each violation type.

Table 1: Worst component-level relative bias observed across the $n = 500$, $B = 200$ sweeps. The three components with the largest $|\text{bias}|/|\text{truth}|$ are shown for each violation, with the coverage of the 95% Wald interval at the same severity. Scale parameters of low-rate components (λ_3 and λ_5 , which have the largest scales hence the lowest hazards) suffer the most catastrophic bias under C3 violation. Numbers are regenerated directly from the sweep RDS files via `paper/regenerate_tables.R`.

Violation	Parameter	Max $ \text{bias} / \text{truth} $	Coverage at worst	Severity
C1	k_5	43%	0.46	$\alpha_{C_1} = 0.50$
	λ_1	37%	0.98	$\alpha_{C_1} = 0.50$
	k_1	30%	0.21	$\alpha_{C_1} = 0.50$
C2	λ_2	93%	0.98	$s = 0.5$
	k_3	37%	0.69	$s = 1.0$
	k_2	23%	0.44	$s = 1.0$
C3	λ_5	163%	0.96	$\alpha_{C_3} = 1.25$
	λ_3	146%	0.99	$\alpha_{C_3} = 1.75$
	λ_2	41%	1.00	$\alpha_{C_3} = 2.00$

Figure 2 shows the RMSE trajectories for all three violations. The system-hazard RMSE is essentially flat across the full severity range, while several component-level RMSE values grow by an order of magnitude.

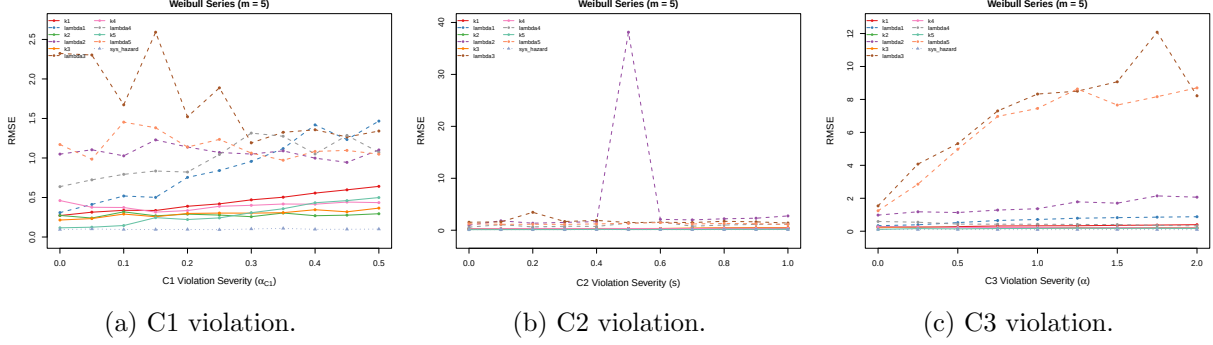


Figure 2: RMSE as a function of violation severity. The system hazard (triangle markers) is stable; component-level RMSEs grow with severity, with scale parameters of low-rate components growing fastest, particularly under C3 violation.

The pattern that emerges across the three sweeps: *scale parameters of components with longer characteristic times (large λ_j , hence low hazard rate) are the most fragile under any masking violation*. These are precisely the components whose contribution to the system hazard is smallest at any given time, so the standard MLE has the least information to identify them. When masking introduces additional uncertainty about which component failed, the scale estimates for low-rate components absorb most of the bias.

4.4 Coverage Anomalies

Figure 3 shows 95% Wald coverage as a function of severity. Several features deserve comment.

Coverage collapse for λ_1 under C3. The most striking coverage failure is for λ_1 under C3 violation: coverage drops from 0.94 at $\alpha_{C_3} = 0$ to 0.00 at $\alpha_{C_3} = 1.0$ and beyond. The bias is large (around -50% relative at the worst point) and the Wald intervals do not widen proportionately, so the truth lies outside every interval. This is the sharpest instance of the general phenomenon: misspecification produces confidently wrong inference at the component level, exactly the failure mode that motivates sensitivity analysis as a prerequisite for reporting component-specific reliability.

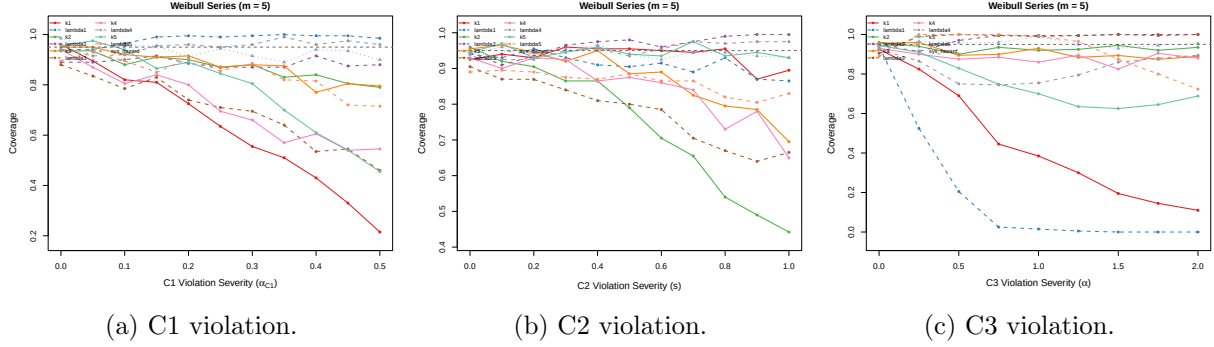


Figure 3: 95% Wald coverage as a function of violation severity. Dashed line at 0.95 nominal. The system hazard maintains near-nominal coverage throughout. Component-level coverage degrades severely under C2 (some parameters drop below 0.40) and collapses to zero for λ_1 at high C3 severity, indicating systematic bias not captured by the Wald interval width.

Coverage inflation for λ_2 under C2. At $s = 0.5$ the λ_2 estimator exhibits an instructive anomaly: its mean bias spikes due to 1–2 replications out of 200 in which the optimizer converges to extreme estimates ($\max \hat{\lambda}_2 = 542$, median 4.56, truth 4.0). The Wald coverage remains near nominal because the estimated standard errors grow commensurately when the Fisher information becomes near-singular for this parameter at the $\mathbf{P}(0.5)$ configuration. This is the dual failure mode: the parameter is effectively unestimable but the confidence interval correctly signals the loss of identifiability through its width.

4.5 The Slope-Test Battery: First-Order Behavior

To test the first-order preservation predictions of Corollary 3.15 formally, we apply a three-test battery to the system-hazard bias trajectories: (1) full-range linear regression, (2) small- α linear regression restricted to the lowest 25% of the sweep range, and (3) a quadratic regression whose linear coefficient isolates the first-order behavior from higher-order curvature.

Table 2 reports the quadratic linear-term test and p -value for each violation at the main sample size $n = 500$. The linear term measures the first-order coefficient $\partial S^\dagger / \partial \alpha|_{\alpha=0}$ that the theory predicts to be zero under first-order preservation.

The empirical results match the theoretical predictions exactly. C2 and C3 violations leave the first-order behavior intact (linear coefficient not significantly different from zero), while C1 violation breaks first-order preservation with a real coefficient of magnitude ~ 0.32 per unit severity for this 5-component Weibull system.

Table 2: First-order behavior test: linear coefficient (with quadratic adjustment) for the system-hazard bias as a function of violation severity, $n = 500$, $B = 200$. A nonzero linear coefficient at the standard significance level indicates failure of first-order preservation.

Violation	Quadratic linear term	p -value	First-order verdict	Theory match
C1	-0.32	< 0.001	broken	matches Corollary 3.15 item 3
C2	-0.04	0.256	preserved	matches Proposition 3.5
C3	+0.004	0.689	preserved	matches Corollary 3.15 item 3

4.6 Sample-Size Ablation ($n = 2000$)

To confirm that the robustness-hierarchy verdicts are not artifacts of finite-sample bias at $n = 500$, we repeat each sweep at $n = 2000$ with $B = 100$ replications. Table 3 reports the slope-test battery for both sample sizes.

Table 3: Slope-test battery across two sample sizes. The quadratic linear term (the first-order coefficient) is qualitatively stable across the $4\times$ sample-size increase: C1 retains its significantly nonzero coefficient (the failure of first-order preservation is structural), while C2 and C3 remain not significantly different from zero (first-order preservation is confirmed). Magnitudes shift slightly because the finite-sample MLE bias at $\alpha = 0$ is smaller at larger n .

Violation	$n = 500, B = 200$		$n = 2000, B = 100$	
	Linear coef.	p	Linear coef.	p
C1	-0.32	< 0.001	-0.30	< 0.001
C2	-0.04	0.256	-0.01	0.305
C3	+0.004	0.689	-0.008	0.204

Three observations from the ablation:

1. **C1 first-order coefficient is structural.** The linear coefficient is essentially identical at the two sample sizes (-0.32 vs. -0.30 , both at $p < 0.001$). The failure of first-order preservation is not finite-sample noise but a real structural feature of the Weibull C1 misspecification.
2. **C2 and C3 first-order preservation is robust.** The linear coefficients remain not significantly different from zero at $p > 0.20$ at both sample sizes.
3. **Component-level fragility is also structural.** The worst component biases at $n = 2000$ are within a few percentage points of those at $n = 500$ (e.g., λ_5 under C3: $+163\%$ at $n = 500$ vs. $+149\%$ at $n = 2000$). The catastrophic component-level errors are a feature of the misspecification, not the sample size.

The detailed $n = 2000$ figures (analogues of Figures 1 to 3) are produced by the ablation scripts in `paper/` and stored alongside the main sweep figures in `inst/simulations/figures/`.

4.7 Practical Guidance

The simulation results yield clean practitioner-facing guidance:

System-level reporting. Reliability engineers reporting system-level estimates (total hazard, MTTF, system survival at fixed times) from the standard masked-data Weibull MLE have substantial protection against any of the three masking-condition violations. The bias is bounded under 4–5% even at severe violation levels and effectively zero under C2 or C3 violations. No correction or sensitivity analysis is required for system-level estimands in typical engineering applications.

Component-level reporting. Component-level estimates (individual rates, shape-scale pairs, component reliability at fixed times) must be reported with explicit sensitivity analysis. The standard model’s worst-case relative bias for component-level scale parameters can exceed 100% under C3 violation, and confidence intervals can have near-zero coverage. Practitioners should compute the robustness intervals of Section 3.8 for each component-level claim before publication.

Diagnostic indicator. Within-system component heterogeneity is the primary risk factor for component-level fragility. The largest biases concentrate on scale parameters of components with longer characteristic times (i.e., lowest hazard rates). Engineers analyzing systems with substantial rate heterogeneity should be especially cautious about component-level inference.

5 Application: Guo et al. Turbine Engine Data

To illustrate the sensitivity analysis workflow, we apply it to the masked failure data from Guo et al. [10], a $m = 3$ component series system with $n = 30$ uncensored failures. Each observation records the system failure time and a candidate set indicating which components could have caused the failure. This dataset has been used as a benchmark in several masked data studies [10, 26].

Standard analysis. Fitting the C1-C2-C3 Weibull series MLE yields estimated shapes $\hat{\mathbf{k}} = (1.26, 1.17, 1.13)$ and scales $\hat{\boldsymbol{\lambda}} = (1000, 897, 847)$. The system hazard at the median failure time $t_0 = 249.5$ is $\hat{h}_T(t_0) = 0.00307$. The large scale estimates reflect the long lifetimes in this dataset; the shape estimates near 1 suggest approximately exponential behavior.

Sensitivity sweep. We construct a family of $\mathbf{P}$ matrices using the same severity parameterization as in Section 4, with a 3×3 cyclic direction matrix and base probability $p_0 = 0.5$. For each severity level, we refit the Weibull series MLE under the relaxed C2 model with known $\mathbf{P}(s)$.

Table 4: Sensitivity analysis of the Guo et al. dataset ($m = 3$, $n = 30$). The system hazard $\hat{h}_T(t_0)$ at $t_0 = 249.5$ is stable across all severity levels; individual scale parameters shift by up to 22%.

s	$\hat{k}_1$	$\hat{k}_2$	$\hat{k}_3$	$\hat{\lambda}_1$	$\hat{\lambda}_2$	$\hat{\lambda}_3$	$\hat{h}_T(t_0)$
0.0	1.26	1.17	1.13	1000	897	847	0.00307
0.3	1.36	1.12	1.08	883	923	965	0.00302
0.5	1.34	1.10	1.15	1032	918	800	0.00308
0.8	1.35	1.08	1.16	1036	933	786	0.00306
1.0	1.36	1.07	1.16	1033	945	780	0.00305

Table 4 confirms the pattern from the simulation study on real data: the system hazard is remarkably stable (varying by less than 2% across the full severity range), while individual component parameters shift with severity. The scale parameter $\hat{\lambda}_3$ decreases from 847 to 780 (an 8% shift) as severity increases, while $\hat{\lambda}_2$ increases from 897 to 945 (a 5% shift). Shape parameters show smaller absolute changes but relative shifts of up to 9%.

For this dataset, a practitioner can confidently report the system-level conclusion—that the system hazard at the median failure time is approximately 0.003—without concern about C2 violations. Individual component rankings, however, are less stable: at $s = 0$, component 3 appears least reliable ($\hat{\lambda}_3 = 847$, smallest scale), but by $s = 1.0$ component 3 has $\hat{\lambda}_3 = 780$ while component 1 has $\hat{\lambda}_1 = 1033$, amplifying the gap. Whether this shift matters depends on how the component estimates will be used.

6 Discussion

The theoretical results of Section 3 and the empirical sweeps of Section 4 together establish a robustness hierarchy for masked series-system reliability inference. We summarize the practical

implications, structural reasons behind the hierarchy, limitations of the analysis, and natural extensions.

6.1 When Does Each Violation Matter?

The robustness hierarchy of Corollary 3.15 partitions estimands into three regimes:

System-level estimands are reliable across all three violations. For the total system hazard, mean time to failure, and system reliability at a fixed time, the standard C1-C2-C3 MLE produces estimates whose bias remains bounded across the full range of practical violation severities. For exponential components, the preservation is exact under any masking violation (Theorem 3.6); the profile-likelihood factorization decouples the system-rate score from the candidate-set information entirely. For Weibull components, preservation is first-order under C2 and C3 violations and admits a small bounded first-order bias under C1 violations. The empirical bound at the 5-component system studied here is under 4% relative bias at $\alpha_{C_1} = 0.5$, and under 1% under C2 or C3.

Component-level estimands are fragile under any violation. Individual component parameters (rates for exponential, shape-scale pairs for Weibull) are not first-order robust to any of the three violations. The empirical results document the failure mode in stark detail: component-level relative biases can exceed 100% (e.g., λ_5 under C3 violation), confidence-interval coverage can collapse from nominal to zero (e.g., λ_1 under C3), and the Fisher information for specific parameter combinations can become near-singular under C2 perturbations (the λ_2 instability at $s = 0.5$). The standard MLE makes confident but systematically wrong claims about which components are unreliable.

The asymmetry between C1 and the other two violations for Weibull. The simulation sweep cleanly separates the three violations by their first-order behavior. Under C2 and C3, the linear coefficient in the pseudo-true expansion is statistically indistinguishable from zero at $p > 0.20$ at both $n = 500$ and $n = 2000$, confirming first-order preservation. Under C1, the linear coefficient is consistent and significantly nonzero (~ -0.30 at both sample sizes, $p < 0.001$). The structural reason traces to where each violation inserts bias into the likelihood: C2 and C3 distort only the candidate-set portion (which couples to the time portion at second order through the Weibull

shape-scale interaction), while C1 violation introduces empty or wrong-cause candidate sets that bias the candidate-set portion more aggressively, and that bias propagates into the time-portion estimates at first order through the shared parameterization.

6.2 Practical Guidance

Since the masking mechanism is a diagnostic process rarely known in detail, the practitioner usually cannot verify whether C1, C2, or C3 holds. The robustness hierarchy yields a simple decision procedure.

1. **Fit the standard model.** The C1-C2-C3 MLE is well-understood, computationally straightforward, and yields reliable system-level estimates regardless of coarsening condition status under the regimes characterized in this paper.
2. **Identify the inferential target.** If only system-level quantities are needed (system hazard, MTTF, system survival at a fixed time), the standard model suffices. No correction or sensitivity analysis is required for exponential components, and only mild caution is warranted for Weibull components under suspected C1 violation.
3. **For component-level inference, perform sensitivity analysis.** Construct a family of plausible masking models representing different violation directions and severities. Refit the standard model under each scenario or use the local sensitivity indices of Section 3.8 to bound the first-order bias. If the component-level conclusions are stable across the family, the violations are not a concern for the application at hand.
4. **Report sensitivity bounds, not point estimates, when conclusions are violation-sensitive.** For component-level claims that move under the assumed masking structure, report the range of estimates as a sensitivity bound. This parallels the partial identification approach [19]: data identify a set of possible component parameter values rather than a point estimate, and reporting the set is more honest than reporting a single estimate under an unverifiable assumption.

6.3 Limitations

Several aspects of the analysis warrant explicit caveats.

- **Parametric perturbation families.** The Bernoulli C1, directional-P C2, and power-weight C3 sweeps are specific parametric violations. Each was chosen to be interpretable and cover a calibrated severity scale, but other violation structures (e.g., masking that depends explicitly on t , diagnostic procedures that interact with system age, masking that depends on observed covariates) are not covered. The first-order preservation arguments of Section 3.3 and Section 3.7 extend in principle to any smooth violation family, but the empirical magnitudes depend on the specific perturbation.
- **Lifetime distributions.** We study Weibull components, with exponential as a constant-hazard special case. Other parametric families (log-normal, Gompertz, gamma, generalized gamma) have different hazard shapes and may exhibit different first-order behavior under the same violations. The exponential exact-preservation result (Theorem 3.6) depends specifically on the additive factorization $\ell(S, \phi) = \ell_S(S) + \ell_\phi(\phi)$ that does not generalize to non-constant-hazard models.
- **Direction dependence for C2.** The specific direction matrix $\mathbf{D}$ in the C2 perturbation determines which component parameters absorb the masking asymmetry. Different $\mathbf{D}$ matrices produce different component-level bias patterns. The system-hazard robustness persists by Proposition 3.5 regardless of $\mathbf{D}$, but component-level fragility claims should be interpreted as representative rather than universal.
- **Sample size.** The main simulations use $n = 500$ with $n = 2000$ as a robustness check. The bias trajectories are qualitatively stable across the $4\times$ sample-size change, but coverage degradation is more pronounced at larger n (confidence intervals narrow while bias persists), an effect already visible in the λ_1 coverage trajectory under C3.
- **Variance estimation.** Coverage uses Wald intervals based on model-based Fisher information, which assumes correct specification. Under masking-condition violations, the sandwich estimator [15, 35] is more appropriate. The observed coverage degradation conflates bias and

incorrect standard errors; disentangling them via sandwich-based coverage is straightforward to add and is left for the software extension.

- **Set-valued versus point-cause masking.** The framework and results apply to set-valued candidate-set masking. As Appendix C shows, this observational structure does not reduce to point-cause misclassification with random coarsening, so the methodology of Ebrahimi [8], Van Rompaye et al. [33], Moreno-Betancur et al. [20], and Bakoyannis et al. [2] is not directly applicable.

6.4 Future Work

Three natural extensions are visible from the current results.

A formal specification test for $\{C_1, C_2, C_3\}$. Under $H_0: \{C_1, C_2, C_3\}$, the conditional distribution of the candidate set given the failure time factors as $\Pr(C = c \mid T = t) = \pi_c \cdot R_c(t; \boldsymbol{\theta}^*)$ where π_c depends only on c and $R_c(t; \boldsymbol{\theta}) = \sum_{k \in c} h_k(t; \theta_k) / S(t; \boldsymbol{\theta})$ is the relative-hazard contribution. This factorization is testable from observed data alone (no need to observe the failed cause K). A bin-based Pearson statistic on candidate-set frequencies provides a workable null distribution; the practical challenge is power against the C1 violation (the most inferentially consequential) at typical reliability-study sample sizes. We have implemented and validated the singleton version of the test in our companion software; substantial development is needed for a publication-ready version, including the all-candidate-set extension and a closed-form sandwich variance.

Second-order theory for the C1 first-order failure. The simulation results show a quadratic-in- α trajectory for the Weibull C1 bias, with a significant quadratic curvature at both sample sizes. Deriving the second-order coefficient $\Delta_j^{(2)}(\boldsymbol{\theta}^*; p_0)$ would explain the bias-bounding phenomenon (why the bias plateaus rather than growing linearly) and provide tighter robustness intervals than the first-order ISNI alone can deliver. The derivation should follow from extending the proof of Theorem 3.9 to the second-order expansion of the score equations.

Extension to other parametric families. The exponential-versus-Weibull asymmetry (exact preservation versus first-order preservation) follows from the constant-hazard property of the

exponential. Parametric families with related structural properties (gamma at integer shape, log-logistic with specific shape constraints, generalized exponential subfamilies) may admit analogous exact preservation results. A systematic survey of which masked-data parametric families admit which preservation tier would strengthen the framework’s generality.

7 Conclusion

We have studied the sensitivity of maximum-likelihood estimation in masked series systems to violations of the three coarsening conditions C1, C2, and C3. The contributions of the paper are unified by the robustness hierarchy that emerges from the analysis:

1. **Calibrated severity scales for each of the three coarsening violations.** Each violation is parameterized on a scalar severity $\alpha_{C_k} \in [0, \alpha_{C_k}^{\max}]$ with $\alpha_{C_k} = 0$ recovering the standard model and $\alpha_{C_k}^{\max}$ marking the boundary to adversarial regimes. Together they form a multi-dimensional sensitivity geometry that complements the calibrated-robustness-value philosophy of Cinelli and Hazlett [3] and the pattern-mixture sensitivity parameter of Moreno-Betancur et al. [20].
2. **Closed-form local sensitivity indices specializing ISNI to set-valued masking.** The first-order expansions of the misspecified MLE at the standard-model fit are derived in closed form for each violation. For exponential components the system-hazard preservation is exact, not first-order, by a profile-likelihood factorization argument (Theorem 3.6); for Weibull components the expansion specializes the ISNI construction of Troxel et al. [29] and Zhang and Heitjan [37] from interval-censoring to candidate-set coarsening.
3. **Robustness intervals for practitioner-facing estimands.** For each violation type and each reliability functional of interest, the robustness interval gives the range of severities within which the estimate remains within a stated tolerance of the standard-model fit. The implementation (Monte-Carlo and first-order variants) is available in the `mdrelax` R package.
4. **Robustness hierarchy classifying estimands.** System-level estimands (total hazard, MTTF, system reliability) are robust to all three violations: exactly for exponential components, at first order for Weibull under C2 and C3, and with a small bounded first-order bias under

C1 for Weibull. Component-level estimands are fragile under all three violations regardless of component family. The simulation sweeps validate the hierarchy at two sample sizes with identical qualitative conclusions.

5. **Reproducible software.** The methods are implemented in the `mdrelax` R package, which provides data generation, MLE, Fisher information, and robustness-interval computation for all model tiers. None of the closest competing methodology papers release code.

The headline message for the practitioner is sharp. A reliability engineer reporting system-level estimates from the standard masked-data Weibull MLE has substantial protection against any plausible violation of the masking-condition assumptions. A reliability engineer reporting component-level estimates does not, and should perform sensitivity analysis using the methods of Section 3.8 before publication.

All code, simulations, and the manuscript source are available at <https://github.com/queelius/mdrelax>. The k-out-of-m generalization of the masking framework, including a threshold theorem characterizing when individual component parameters are identifiable, is developed in companion work [27].

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A Score Functions

For reference, we provide the score functions under the misspecified and true models for series systems with general component hazards.

Misspecified score (C1-C2-C3). The score under the standard model for parameter θ_j is:

$$\frac{\partial \ell^{\text{wrong}}}{\partial \theta_j} = \sum_{i=1}^n \frac{\partial \log R_j(s_i; \theta_j)}{\partial \theta_j} + \sum_{i:\delta_i=1} \frac{\mathbf{1}_{j \in c_i} h'_j(s_i; \theta_j)}{\sum_{k \in c_i} h_k(s_i; \theta_k)}, \quad (35)$$

where $h'_j = \partial h_j / \partial \theta_j$.

True score (C1-C3 with known P). Under the Bernoulli masking model with masking weights

$\pi_{k,c}$:

$$\frac{\partial \ell^{\text{true}}}{\partial \theta_j} = \sum_{i=1}^n \frac{\partial \log R_j(s_i; \theta_j)}{\partial \theta_j} + \sum_{i:\delta_i=1} \frac{\pi_{j,c_i} \mathbf{1}_{j \in c_i} h'_j(s_i; \theta_j)}{\sum_{k \in c_i} h_k(s_i; \theta_k) \pi_{k,c_i}}. \quad (36)$$

Exponential specialization. For exponential components ($h_j = \theta_j$, $R_j = e^{-\theta_j t}$), these simplify

to:

$$\frac{\partial \ell^{\text{wrong}}}{\partial \theta_j} = - \sum_{i=1}^n s_i + \sum_{i:\delta_i=1} \frac{\mathbf{1}_{j \in c_i}}{\sum_{k \in c_i} \theta_k}, \quad (37)$$

$$\frac{\partial \ell^{\text{true}}}{\partial \theta_j} = - \sum_{i=1}^n s_i + \sum_{i:\delta_i=1} \frac{\pi_{j,c_i} \mathbf{1}_{j \in c_i}}{\sum_{k \in c_i} \theta_k \pi_{k,c_i}}. \quad (38)$$

B Non-Identifiability Evidence

To demonstrate the non-identifiability of $(\boldsymbol{\theta}, \mathbf{P})$, we attempted joint estimation with $n = 2000$ observations from a 2-component exponential system with $\boldsymbol{\theta}^* = (1.0, 2.0)$ and asymmetric $\mathbf{P}$ ($p_{21} = 0.10$, $p_{12} = 0.90$).

Method	$\hat{\theta}_1$	$\hat{\theta}_2$
Joint $(\boldsymbol{\theta}, \mathbf{P})$	1.62	1.44
Known $\mathbf{P}$	0.97	2.08
True values	1.00	2.00

The joint estimator consistently converges to $\hat{\boldsymbol{\theta}} \approx (1.6, 1.4)$ with $\hat{P}_{21} \approx 0.45$ (true: 0.10), regardless of sample size. The total hazard is well-identified: $\sum_j \hat{\theta}_j = 3.01$ vs. true 3.00. This confirms Theorem 3.8: individual rates and masking probabilities are confounded, but the total system hazard is identifiable.

C Set-Valued Masking versus Point-Cause Misclassification

A natural question for readers familiar with the biostatistics competing-risks literature is whether the relaxed-C1 likelihood developed here can be obtained as a special case of point-cause misclassification with random masking. Methods for the latter setting, including Ebrahimi [8], Van Rompaye et al. [33], Ha and Tsodikov [12], and Bakoyannis et al. [2], might then apply directly. We show that the two model families are non-nested and that set-valued masking under C1 violation cannot, in general, be obtained from point-cause misclassification composed with random coarsening.

Definition C.1 (Point-cause misclassification with random masking). *The data-generating process is:*

1. The true failed component $K \in \{1, \dots, m\}$ is drawn from the system-failure distribution.
2. A recorded cause $K^{\text{obs}} \in \{1, \dots, m\}$ is drawn from a misclassification distribution $\pi_{j|k} = \Pr\{K^{\text{obs}} = j \mid K = k\}$.
3. The candidate set $C \subseteq \{1, \dots, m\}$ is drawn conditional on $K^{\text{obs}} = j$ from some distribution $\rho(c \mid j)$ satisfying $\rho(c \mid j) > 0$ only if $j \in c$.

Lemma C.2 (Non-nesting of set-valued masking and point-cause models). *Let $\mathcal{P}_{\text{set}}(m)$ denote the family of joint distributions of (K, C) generated by Bernoulli set-valued masking under relaxed C1, in which conditional on $K = k$ the candidate set C is formed by including each component j independently with probability $q_j(k) \in [0, 1]$. Let $\mathcal{P}_{\text{point}}(m)$ denote the family generated by Definition C.1. Then for $m \geq 2$, $\mathcal{P}_{\text{set}}(m)$ and $\mathcal{P}_{\text{point}}(m)$ are non-nested: neither family is a subset of the other.*

Proof. $\mathcal{P}_{\text{set}} \not\subseteq \mathcal{P}_{\text{point}}$: Choose $q_j(k) \in (0, 1)$ for all j, k . Then

$$\Pr\{C = \emptyset \mid K = k\} = \prod_{j=1}^m (1 - q_j(k)) > 0.$$

Under $\mathcal{P}_{\text{point}}$, the candidate set C contains K^{obs} almost surely, so $C \neq \emptyset$ almost surely.

$\mathcal{P}_{\text{point}} \not\subseteq \mathcal{P}_{\text{set}}$: For $m = 4$ and any K , take $\pi_{2|1} = \pi_{3|1} = 1/2$, $\rho(\{2, 4\} \mid 2) = 1$, $\rho(\{3, 4\} \mid 3) = 1$.

Then conditional on $K = 1$:

$$\Pr\{C = \{2, 4\} \mid K = 1\} = 1/2,$$

$$\Pr\{C = \{3, 4\} \mid K = 1\} = 1/2,$$

$$\Pr\{2 \in C, 3 \in C \mid K = 1\} = 0.$$

Under $\mathcal{P}_{\text{set}}$ with marginal inclusion probabilities $q_2(1) = \Pr\{2 \in C \mid K = 1\} = 1/2$ and $q_3(1) = 1/2$, independence of inclusion implies

$$\Pr\{2 \in C, 3 \in C \mid K = 1\} = q_2(1) q_3(1) = 1/4 \neq 0,$$

contradicting the joint distribution above. □

Remark C.1 (Consequences for inference). Direction (i) of Lemma C.2 shows that relaxed-C1 distributions cannot be recovered from point-cause misclassification models without explicitly admitting an “empty candidate set” outcome that is not part of those models’ standard formulations. Direction (ii) shows that point-cause misclassification with arbitrary masking conditional on the recorded cause can produce correlation structures in candidate-set inclusion that the Bernoulli set-valued model cannot reproduce.

Practically, this means that estimators and sensitivity-analysis tools developed for cause-of-failure misclassification, in particular those of Ebrahimi [8], Van Rompaye et al. [33], Ha and Tsodikov [12], Moreno-Betancur et al. [20], and Bakoyannis et al. [2], are not directly applicable to the set-valued candidate-set setting studied in this paper. Conversely, the relaxed-C1 analysis here does not subsume those methods for the point-cause setting. The two literatures address related but distinct observational structures.

D Software

All methods are implemented in the `mdrelax` R package (version 1.0.0), available at <https://github.com/queelius/mdrelax>. The package provides:

- Data generation under C1-C2-C3 and relaxed C2 for both exponential and Weibull series systems.
- MLE with analytical score and Fisher information for all model tiers.
- The $\mathbf{P}$ matrix construction and masking probability computation functions used in this paper.
- 1276 unit tests verifying likelihood correctness, score–gradient agreement, FIM consistency, and MLE convergence properties.

The sensitivity sweep (`paper/run_sensitivity_sweep.R`) reproduces all figures and tables in Section 4.